

4/8/26

2.

NETWORKING

Day - 1 Number System

- ↳ Types:
- 1) Binary $\rightarrow 0, 1$
 - 2) Decimal $\rightarrow 0-9$
 - 3) Hexa $\rightarrow 0-15$

Base Value

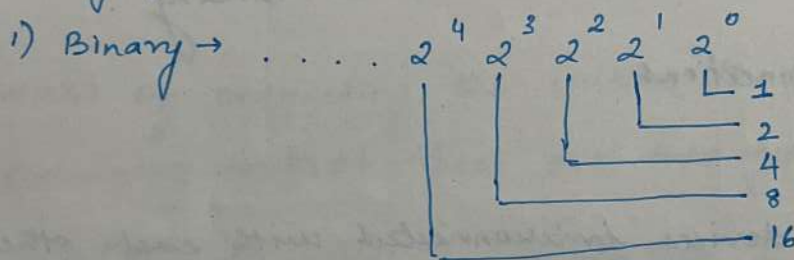
2

10

16

1
2
3
:
10 - A
11 - B
12 - C
13 - D
14 - E
15 - F

↳ Finding place values:



[Base value - 2
Increase the power
left to right]

Place Value
of Binary
(X)

↳ Conversions:

1) Decimal to Binary

[1 octet $\rightarrow$ 8 bit]

128	64	32	16	8	4	2	1
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$\rightarrow$ powers of 2

12 :	0	0	0	0	1	1	0	0
20 :	0	0	0	1	0	1	0	0
128 :	1	0	0	0	0	0	0	0
192 :	1	1	0	0	0	0	0	0
224 :	1	1	1	0	0	0	0	0
240 :	1	1	1	1	0	0	0	0
248 :	1	1	1	1	1	0	0	0
255 :	1	1	1	1	1	1	1	1

2) Hexa $\rightarrow$ Binary

1 hexa = 4 bit

	8	4	2	1
A	1	0	1	0
B	1	0	1	1
C	1	1	0	0
D	1	1	0	1
E	1	1	1	0

IP $\rightarrow$ Decimal format [Number]

MAC $\rightarrow$ Hexadecimal format [Alphabet + Number]

* Computer understands binary nos. only
So, decimal & hexa are converted to binary.